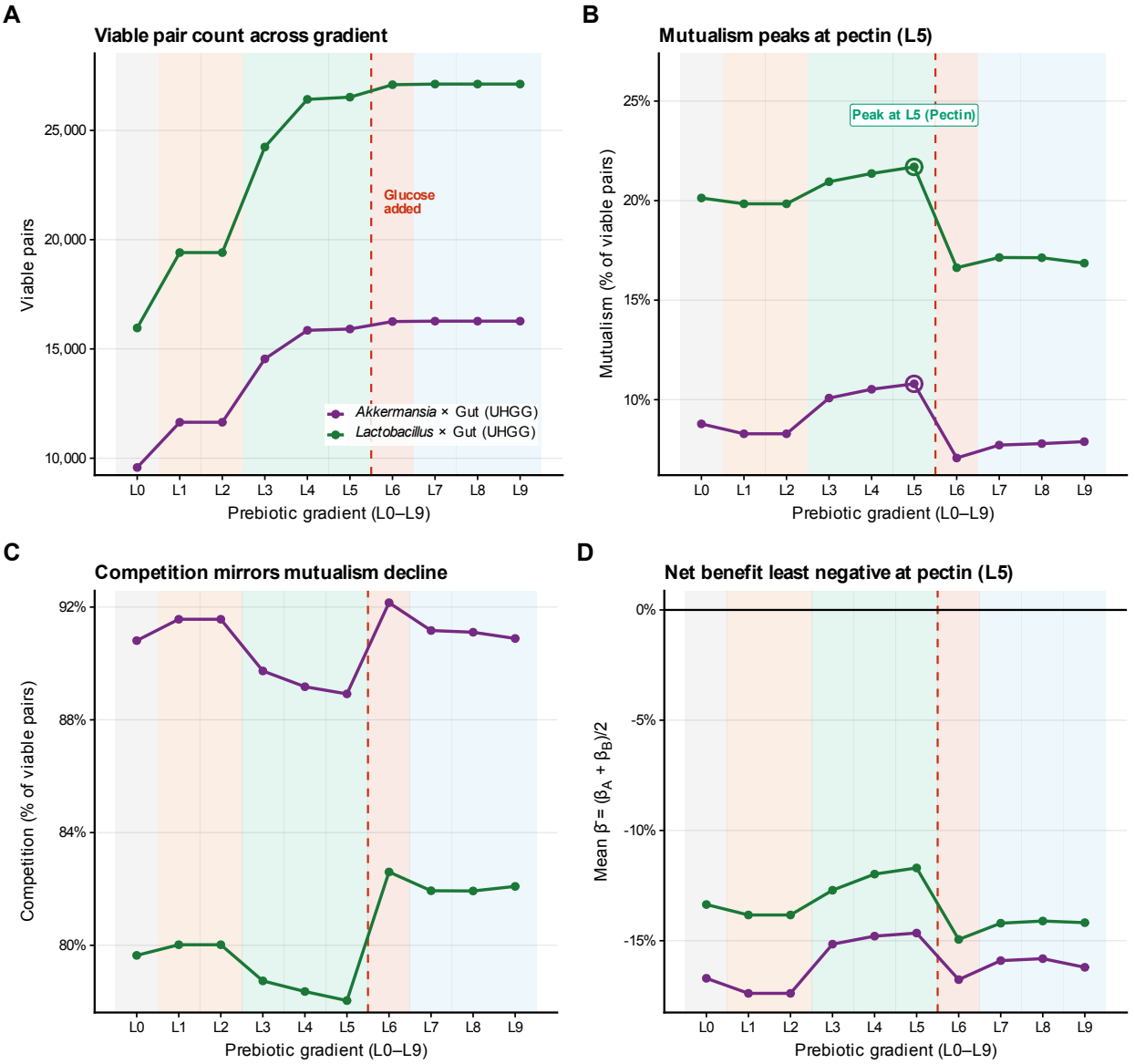

A
Viable pair count across gradient
Viable pairs
25,000
20,000
15,000
10,000
Glucose added
Akkermansia × Gut (UHGG)
Lactobacillus × Gut (UHGG)
L0 L1 L2 L3 L4 L5 L6 L7 L8 L9
Prebiotic gradient (L0–L9)
B
Mutualism peaks at pectin (L5)
Mutualism (% of viable pairs)
25%
20%
15%
10%
Peak at L5 (Pectin)
L0 L1 L2 L3 L4 L5 L6 L7 L8 L9
Prebiotic gradient (L0–L9)
C
Competition mirrors mutualism decline
Competition (% of viable pairs)
92%
88%
84%
80%
L0 L1 L2 L3 L4 L5 L6 L7 L8 L9
Prebiotic gradient (L0–L9)
D
Net benefit least negative at pectin (L5)
Mean β = (β_A + β_B)/2
0%
-5%
-10%
-15%
L0 L1 L2 L3 L4 L5 L6 L7 L8 L9
Prebiotic gradient (L0–L9)